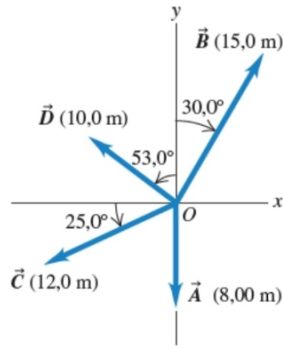
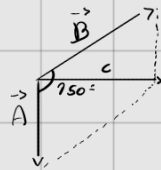


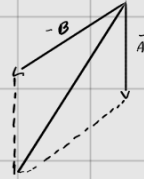
Figura E1.24



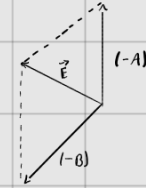
$A = B_m \quad B = 15_m$
 1.29) a) $\vec{A} + \vec{B} = \vec{C}$



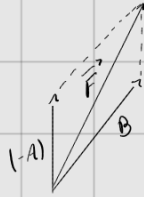
b) $\vec{A} + (-\vec{B}) = \vec{D}$



c) $(-\vec{A}) + (-\vec{B}) = \vec{E}$



d) $\vec{B} + (-\vec{A}) = \vec{F}$



1.28) $\tan \theta = \frac{A_y}{A_x}$

$\theta = \arctg \frac{A_y}{A_x}$

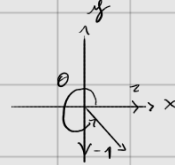
a) $\tan \theta = -\frac{1}{2} \quad A_y = -1$

$\theta = \arctg(-\frac{1}{2}) \quad A_x = 2$

360°

$\theta \approx 333^\circ$

* 4º quadrante

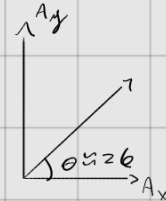


b) $\tan \theta(\frac{1}{2}) \quad A_x = 2$

$\theta = \arctg(\frac{1}{2}) \quad A_y = 1$

$\theta \approx 26^\circ$

* Primeiro quadrante

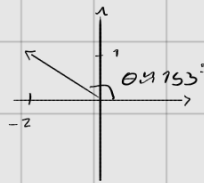


c) $\tan(-\frac{1}{2}) \quad A_x = -2$

$\alpha = \arctg(-\frac{1}{2}) \quad A_y = 1$

$\alpha \approx -26^\circ$

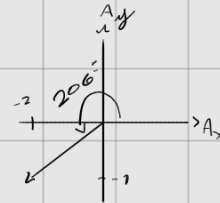
$180 - 26 \approx 153^\circ$



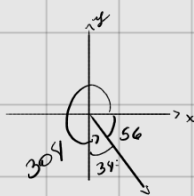
d) $\tan(-\frac{1}{2})$

$\gamma = \arctg(\frac{1}{2})$

$\gamma = 180 + 26 \approx 206^\circ$



1.30) $A_x = -16_m$



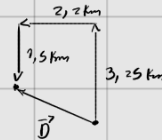
$\tan 56 = \frac{A_y}{-16}$

$\Delta y = -23$

$|A| = \sqrt{A_x^2 + A_y^2}$

$|A| \approx 28,59$

1.33)



1.36) a) $x = 5 \quad y = -6,3$ b) $x = -9,9 \quad y = 11,2$ c) $x = -15 \quad y = 22,4$ d) $\vec{A} = 5\vec{i} \quad \vec{B} = 4\vec{i} - 6\vec{j} \quad A = 20\vec{i} - 30\vec{j}$

1.39) a) $A_y = A \cdot \sin 70 \rightarrow A_y = 3,6 \cdot \sin 70 = 3,38 \quad A = 1,23\vec{i} + 3,38\vec{j}$

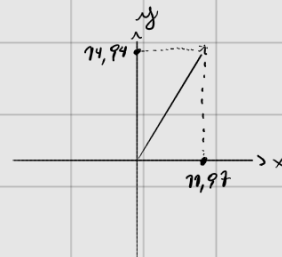
$A_x = A \cdot \cos 70 \rightarrow A_x = 3,6 \cdot \cos 70 = 1,23$

b) $B_x = B \cdot \cos 30 \rightarrow B_x = 2,4 \cdot \cos 30 \rightarrow B_x = -2,07 \quad B = 2,07\vec{i} + 1,2\vec{j}$

$B_y = B \cdot \sin 30 \rightarrow B_y = 2,4 \cdot \sin 30 \rightarrow B_y = 1,2$

c) $3\vec{A} = 3,69\vec{i} + 10,14\vec{j} \quad -4\vec{B} = -8,28\vec{i} - 4,8\vec{j}$

$\vec{C} = (3,69\vec{i} + 10,14\vec{j}) - (-8,28\vec{i} - 4,8\vec{j}) \rightarrow \vec{C} = 11,97\vec{i} + 14,94\vec{j}$



1.40) $\vec{A} = -3\vec{i} + 6\vec{j}$

$\vec{B} = 7\vec{i} + 2\vec{j}$

$\vec{C} = 4\vec{i} + 8\vec{j}$

a) O vetor $\vec{A}$ forma -63° em sentido horário com o x

$\tan \theta = \frac{6}{-3} \rightarrow \theta = \arctg(-2) = -63^\circ$

b) O vetor $\vec{B}$ forma $15,9^\circ$ com o eixo x

$\theta = \arctg(\frac{2}{7}) = 15,9^\circ$

c) $\vec{C} = \vec{A} + \vec{B}$

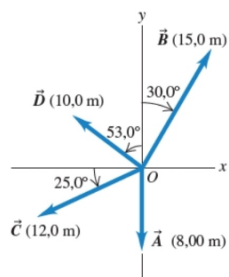
$\vec{C} = 4\vec{i} + 8\vec{j}$

$\theta = \arctg(\frac{8}{4}) = 63,4^\circ$

* O vetor $\vec{C}$ forma $63,4^\circ$ com o eixo x.

1.43)

Figura E1.24



$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta \quad * \text{produto escalar}$$

$$a) \vec{A} \cdot \vec{B} = 8 \cdot 15 \cdot \cos 150^\circ = -104 \text{ m}^2$$

$$b) \vec{B} \cdot \vec{C} = 15 \cdot 12 \cdot \cos 145^\circ = -147 \text{ m}^2$$

$$c) \vec{A} \cdot \vec{C} = 8 \cdot 12 \cdot \cos 65^\circ = 40,6 \text{ m}^2$$

$$1.45) a) \vec{A} = -2\hat{i} + 6\hat{j} \quad \text{e} \quad \vec{B} = 2\hat{i} - 3\hat{j}$$

$$\tan \theta = \frac{6}{-2} = -3$$

$$\theta = -71^\circ$$

$$\tan \alpha = \frac{-3}{2} = -1,5$$

$$\alpha = 56^\circ$$

$\beta \approx 15^\circ$
entre os ângulos

$$b) \vec{A} = 3\hat{i} + 5\hat{j} \quad \text{e} \quad \vec{B} = 7\hat{i} + 6\hat{j}$$

$$\tan \theta = \frac{5}{3}$$

$$\theta = 59^\circ$$

$$\tan \alpha = \frac{6}{7}$$

$$\alpha = 30,9^\circ$$

$\beta \approx 28,1^\circ$
entre os ângulos

$$c) \vec{A} = -4\hat{i} + \hat{j} \quad \text{e} \quad \vec{B} = 4\hat{i} + 7\hat{j}$$

$$\tan \theta = \frac{1}{-4}$$

$$\theta = -26,5^\circ$$

$$\tan \alpha = \frac{7}{4}$$

$$\alpha = 63^\circ$$

$\beta \approx 89,5^\circ$

$$1.46) \vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$|\vec{A}| = 2,8$$

$$\vec{A} \times \vec{B} = 2,8 \cdot 1,9 \cdot \sin 720^\circ$$

$$|\vec{B}| = 1,9$$

$$\vec{A} \times \vec{B} = 4,67 \text{ km}$$

$$1.47) \vec{A} \times \vec{D}$$

$$|\vec{A}| = 8$$

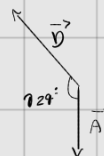
$$|\vec{D}| = 10$$

$$\theta = 233^\circ$$

$$b) \vec{A} \times \vec{D} = 8 \cdot 10 \cdot \sin 72^\circ$$

$$\vec{A} \times \vec{D} = 63,8$$

→ Ângulo entre os dois vetores



$$1.48) \vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$a) \vec{A} \cdot \vec{B} = 3,6 \cdot 2,4 \cdot \cos 140^\circ$$

$$\vec{A} \cdot \vec{B} = -6,62$$

$$b) \vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{A} \times \vec{B} = 3,6 \cdot 2,4 \cdot \sin 140^\circ$$

* Produto apontar para
y negativo.

$\vec{A} \times \vec{B} = 5,5$ * Exz produto vai apontar para y positivo